



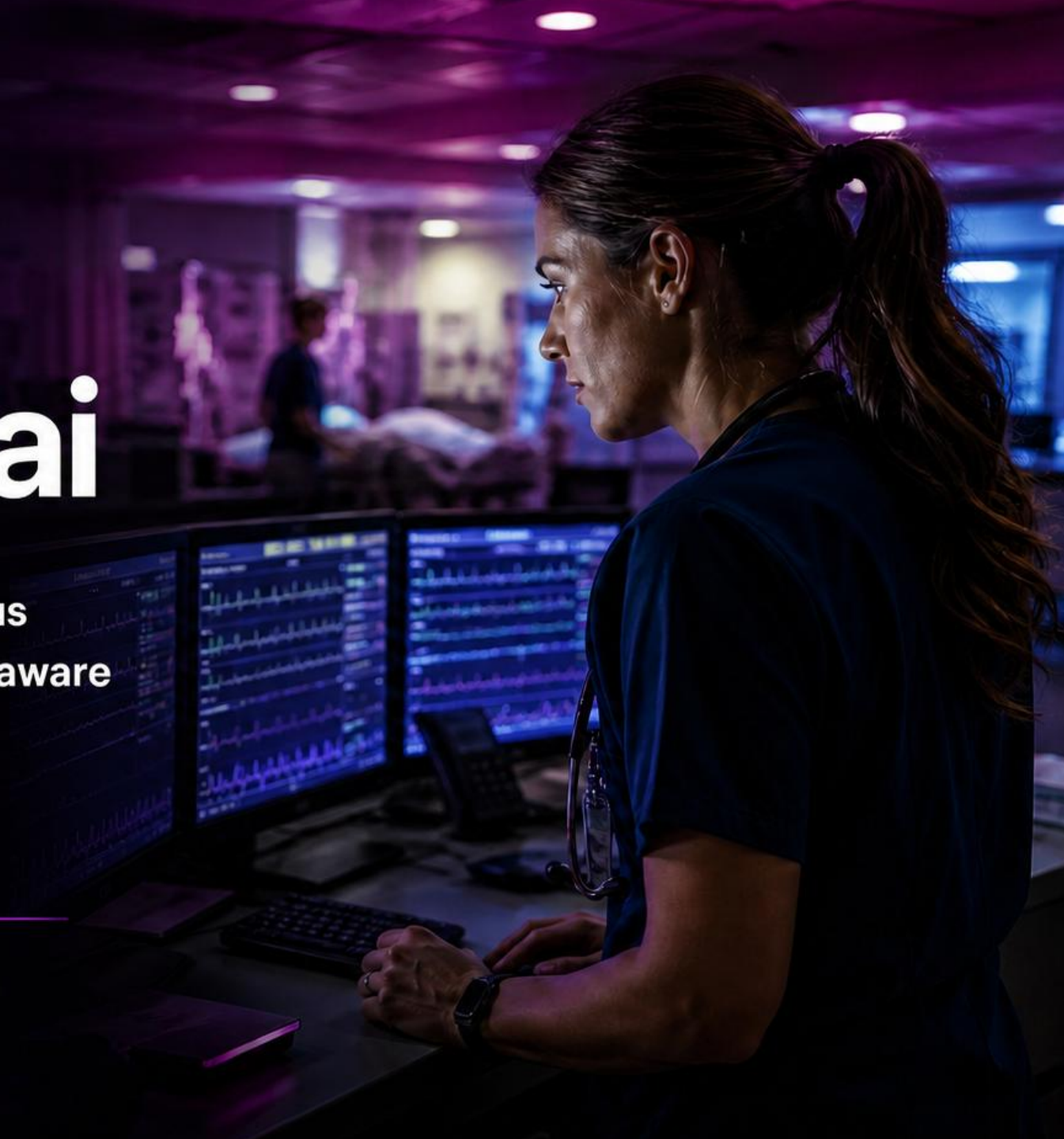
ROUND 2 BUSINESS PROPOSAL

TriageLoop.ai

A clinician-governed AI system for continuous emergency-department triage and capacity-aware prioritisation.



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Team: Zeta



Under 3× demand, 17.7% of patient action windows fail—the unmanaged wait is the real clinical blind spot.



1 | TRIAGE IS A SNAPSHOT

An arrival category records acuity once.



2 | PATIENT RISK KEEPS MOVING

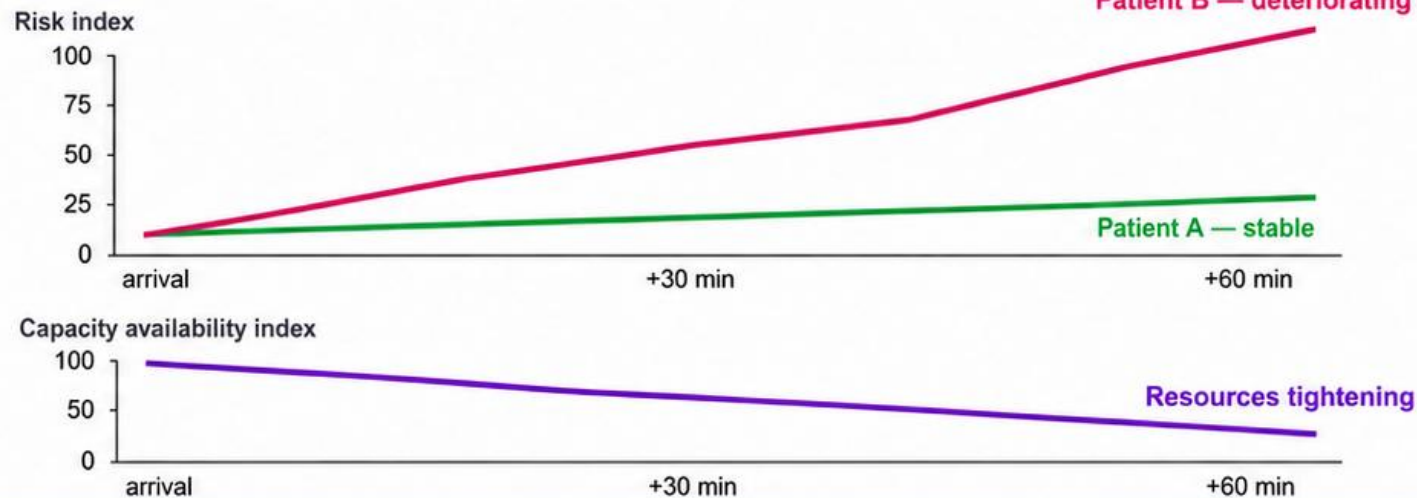
Risk can rise before a clinician sees patient.



3 | CAPACITY MOVES TOO

Beds, staff and flow constraints shift constantly.

Same triage label. Two different trajectories.



17.7%

Action Window misses

30–90 min

unmanaged waiting period

11

capacity conflicts at 3× surge

THE PROBLEM STATEMENT REQUIRES SIX THINGS—TRIAGELOOP COVERS ALL SIX



Continuous monitoring



Age-aware rules



Uncertainty & deferral



Worsening-vitals reassessment



Clinician override



Tamper-evident audit

The category can still look correct while the operating promise is already broken.
TriageLoop tests that promise against a live, adversarially reviewed system — not a slide.

TriageLoop cuts missed windows 20.5% by converting clinical need into time—and time into a capacity decision.



Clinical Slack = Action Window – projected ETA

ONE PATIENT, ONE SIGNAL, ONE ACCOUNTABLE ACTION

Regional ED Waiting room 3x surge | 22 Aug • 10:42 IST | RN - A. Mehta

Capacity constrained 11 patient deadlines are infeasible at current capacity. Escalate operational support.

Live action queue Ranked by clinical level, Action Window and Clinical Slack

12 waiting 6 action due 11 conflicts

All Attention Conflict Safe Mode Find patient

#	Patient	Action now	By	ETA	Slack
1	P-0008 Adult • 63y • 41m feels odd	Immediate review Safe Mode	Now	~4m	~4m infeasible
2	P-0009 Adult • 39y • 35m cough	Immediate review Conflict	Now	~8m	~8m infeasible
3	P-0013 Geriatric • 72y • 31m shortness of breath	Safe Mode review Safe Mode	Now	~12m	~12m infeasible
4	P-0019 Adult • 30y • 28m abdominal pain in pregnancy	Reassess Conflict	55m	~10m	~8m infeasible
5	P-0026 Geriatric • 74y • 25m fever and confusion	Escalate priority Conflict	55m	~14m	~9m infeasible
6	P-0003 Paediatric • 5y • 22m fever	Immediate review Conflict	Now	~16m	~16m infeasible
7	P-0007 Adult • 45y • 19m	Continue monitored wait	515m	~17m	~8m

Model state as of 10:42:00 Live via SSE - polling fallback available

Synthetic decision-support prototype — not validated for clinical use.

Trajectory

Calibrated critical-risk horizons

5m	15m	30m	60m
0%	7%	12%	39%
HR 106 bpm ↑	SpO ₂ 88% ↓	RR 37 /m ↑	BP 108 /70 ↓

Why this action

Current recommendation, not a diagnosis

What changed

- SpO₂ fell to 88%
- respiratory rate rose to 37/min
- heart rate rose to 106 bpm

Factors used

- oxygen saturation (88%) increased estimated urgency
- oxygen-saturation trend (+5 points) increased estimated urgency
- respiratory-rate trend (+7/min) increased estimated urgency

Safety Rule: Determine Tim Action
HARD RED FLAG WORSENING TRAJECTORY; WAIT THRESHOLD - Model Output; Cannot Alter The Rule Outcome

Clinician decision

Human action is recorded against this exact recommendation.

Accept action Modify Override

Model 1.0.0 • Rules 1.0.0 • Recommendation REC-597a1e1da000

BORROWED IDEA → ED OPERATING ADVANTAGE

Recomposition of established techniques, not claimed as individually novel — see closest prior art on the differentiation slide.

Predictive maintenance	Action Window	By when must we act?
Real-time scheduling	Clinical Slack	Can capacity meet it?
Conformal prediction	Safe Mode	When must AI defer?
Digital twins	Queue Twin	What happens under this load?

NON-NEGOTIABLE SAFETY INVARIANTS



ACTION CONTRACT — SAMPLE OUTPUT

Patient: P-0009 ETA: ~8 min
 Action: Immediate review Clinical Slack: ~8 min
 By: NOW Owner: RN A. Mehta

ACCEPT

MODIFY

OVERRIDE

Against fixed 15-minute re-triage, TriageLoop cuts misses 20.5%—with no lower-acuity wait penalty.

PAIRED OPERATIONAL OUTCOMES | LOWER IS BETTER

● Comparator = fixed 15-minute periodic re-triage ● TriageLoop (Dynamic Action Window + Clinical Slack)



100 = comparator baseline; lower is better

CREDIBILITY STACK



1,200

paired synthetic shifts

3 sites × baseline/3× demand



73/73

automated tests passed

contracts, safety, queue, product



95% CI

17.4%–23.8%

paired bootstrap interval



0

hidden negative results

failed NBO gate remains visible

2.8pp — widest recall gap across pediatric / adult / geriatric patients at the safety-critical 30-minute horizon

PAIRED BOOTSTRAP INTERVAL (PRIMARY OUTCOME)



NBO: A PREDECLARED GATE WE FAILED — AND REPORTED



87.5%

fewer measurements

EFFICIENCY GAIN



7.1 pp

below critical-recall gate

SAFETY GAP



GUIDANCE

reassess or escalate

SAFE RELEASE SCOPE

We built the safety check that could kill our own feature. It did. We're showing you anyway.

EVIDENCE PATHWAY: RIGOR BUILT IN, STEP BY STEP

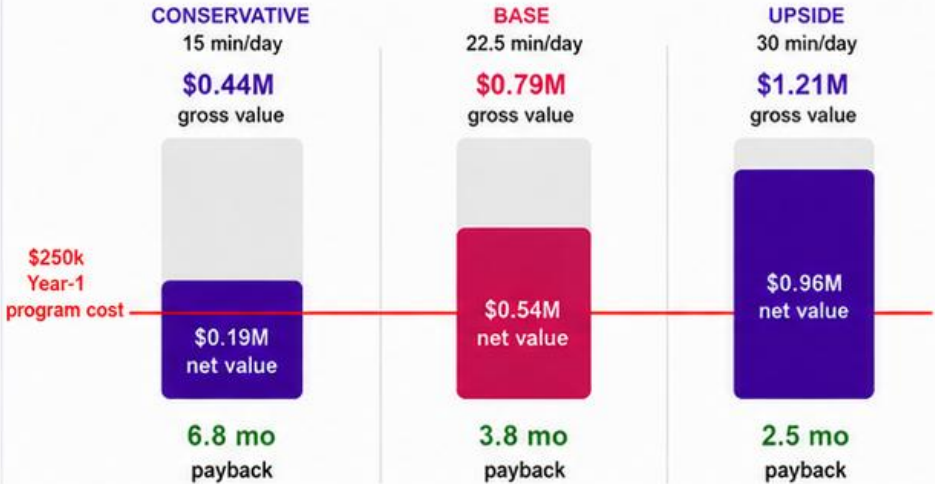


No regression observed in low-acuity P90 wait

The modeled business case breaks even even with only 6–9 attributable minutes of daily boarding improvement.

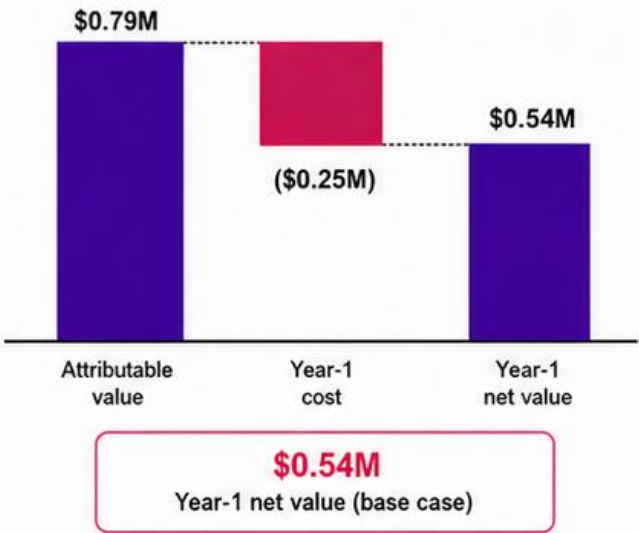
ANNUAL VALUE SENSITIVITY

Illustrative attributable annual value (USD)



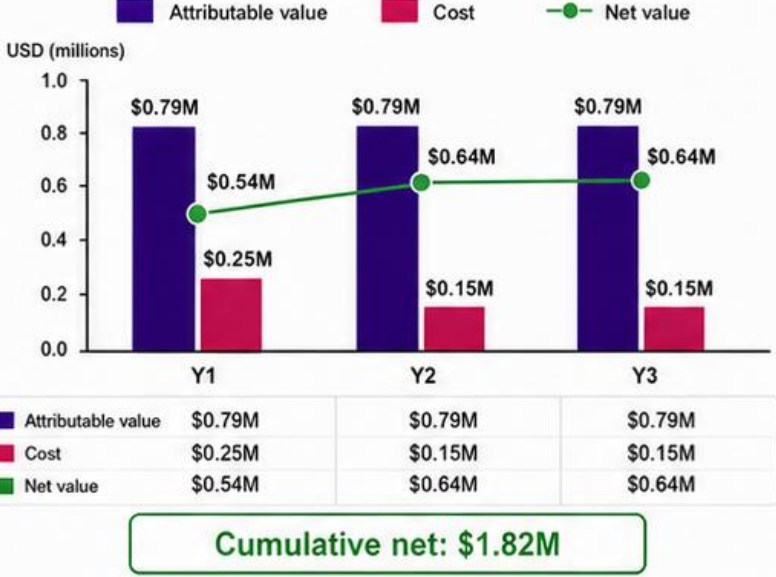
BASE-CASE VALUE BRIDGE

Illustrative Year-1 (USD)



3-YEAR BASE-CASE PROJECTION

Illustrative (USD)



NET VALUE AFTER \$250k COST

Illustrative Year-1 net value (USD millions)

	25% attribution	50% attribution (Base)	75% attribution
15 min/day	-\$0.03M	\$0.19M	\$0.42M
22.5 min/day (Base)	\$0.16M	\$0.54M	\$0.92M
30 min/day	\$0.33M	\$0.96M	\$1.58M

Legend: Negative (<\$0) (Red), Marginal (\$0 to \$0.25M) (Purple), Strong positive (≥\$0.25M) (Green)

VALUE MODEL & KEY ASSUMPTIONS

Illustrative

- Annual attributable value =** boarding improvement × \$/hour × 365 × attribution
- Economic anchor:** \$9,693–\$13,298/day per 1-hour boarding reduction
- Attribution haircut:** 50%
- Year-1 cost assumption:** \$250k

WHAT IS NOT MONETIZED

Illustrative upside not included in this model

- ~~Avoided patient harm~~
- ~~Nurse time released~~
- ~~Audit efficiency~~
- ~~Reduced LWBS~~

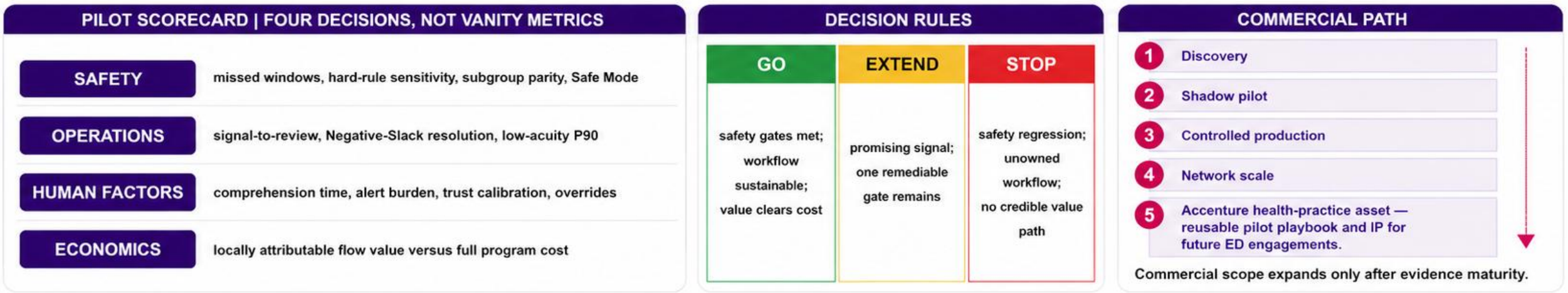
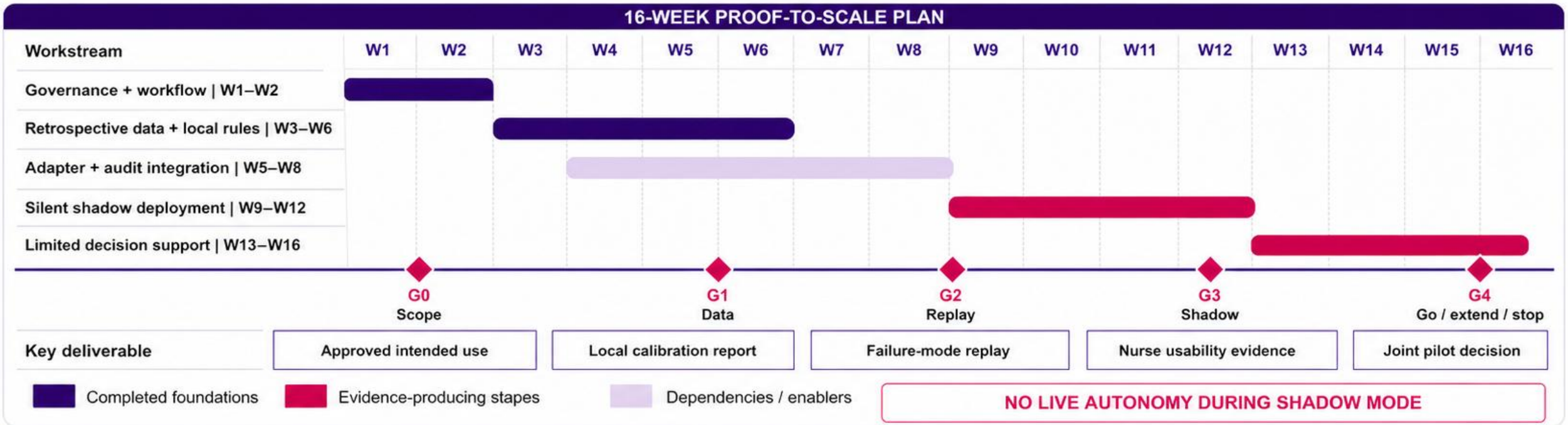
Use as a pilot go/no-go threshold—not a savings promise.

BEYOND ONE SITE (DIRECTIONAL, NOT MODELED)

- India has 190,000+ registered hospital beds in tertiary/secondary EDs across public and private systems
- Single-site economics are the funding gate, not the ceiling — replicable per-site cost structure, not custom-built per hospital
- Commercial path (Slide 11) scales through configuration, not re-engineering.

DIRECTIONAL

A 16-week governed pilot converts synthetic promise into three investment decisions—with zero autonomous clinical action.



07

One accountable action replaces five disconnected alerts—without changing clinical authority.

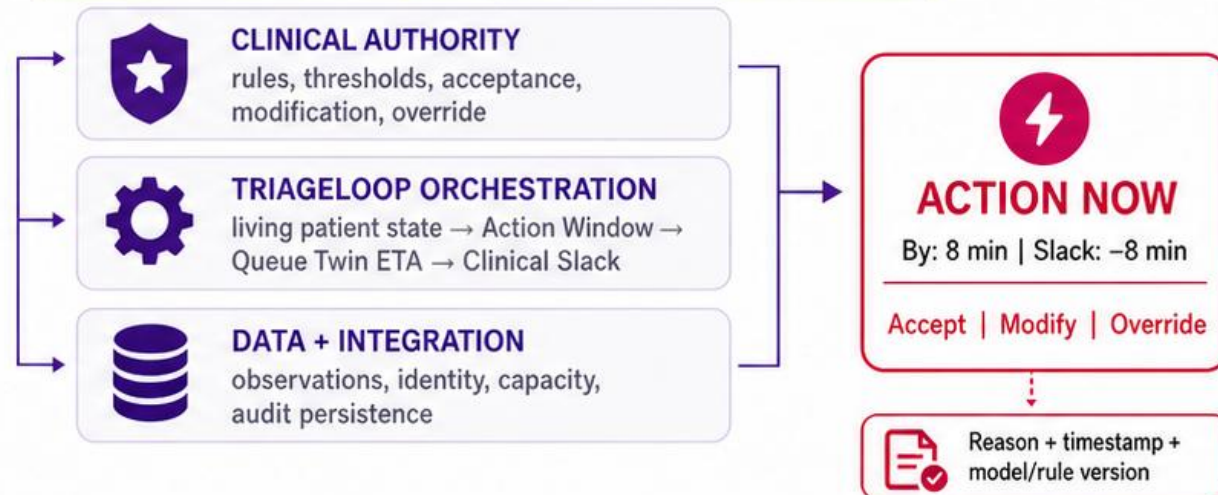
FROM ALERT FRAGMENTATION TO A CLOSED-LOOP OPERATING MODEL

TODAY | FIVE SIGNALS, MANUAL RECONCILIATION



WHAT CHANGES?

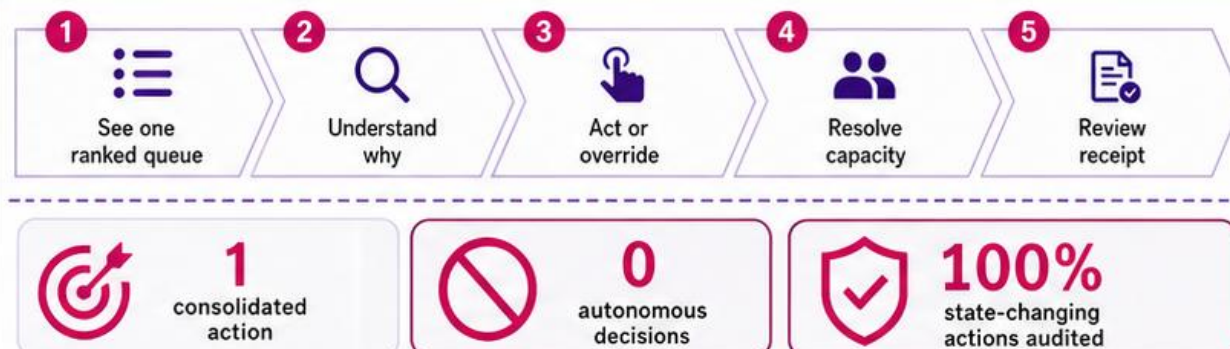
WITH TRIAGELOOP | ONE CONSOLIDATED ACTION



OPERATING OWNERSHIP

	Owns decision	Owns response	Owns assurance
Triage nurse	✓ patient action	—	✓ records reason
Charge nurse	—	✓ capacity conflict	✓ escalation closure
Clinical safety	✓ rule release	✓ incident response	✓ subgroup gates
IT + data	—	✓ integration uptime	✓ audit recovery
Leadership	✓ intended use	✓ resource escalation	✓ attributable value

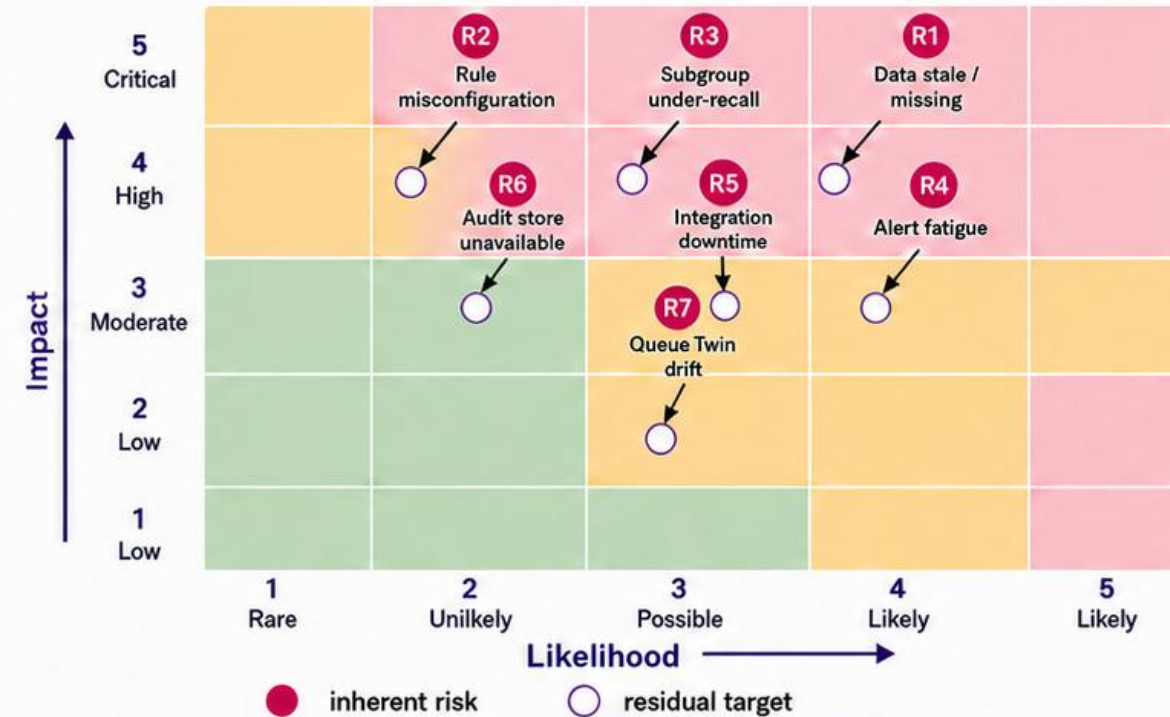
ADOPTION DESIGN | FIT THE WORKFLOW TO THE NURSE



The product coordinates attention; the clinician retains authority.

08 Safety is the operating model: every failure becomes visible, owned and reversible.

CLINICAL + OPERATIONAL RISK MAP



FAILURE-MODE CONTRACT | DETECT → DEGRADE → OWN

Trigger	System response	Human owner	Recovery evidence
Stale / implausible data	Safe Mode: preserve or raise priority	Triage nurse	reviewed observation
Hard red flag	Deterministic rule wins	Clinical safety	rule trace
Model unavailable	Rules + category windows remain	IT + data	service replay
Queue Twin unavailable	Need remains; ETA / Slack unknown	Charge nurse	capacity reconciliation
Audit unavailable	Block state-changing decision	IT + data	write recovery
Uncertain / OOD	Conformal review + Safe Mode	Clinician	deferral receipt
NBO gate not met	Guidance only; reassessment required	Clinical safety	release-gate record

THREE LINES OF ASSURANCE

- Frontline control** nurse accepts, modifies or overrides
- Clinical assurance** rules, thresholds, subgroup gates, incident review
- Technical + governance** monitoring, audit recovery, release approval

RELEASE GATES

- Safety** no hard-rule or subgroup regression
- Workflow** alert burden and comprehension acceptable
- Operations** capacity conflicts visible and owned
- Value** local attributable benefit clears full cost

INCIDENT CONTROL LOOP



09

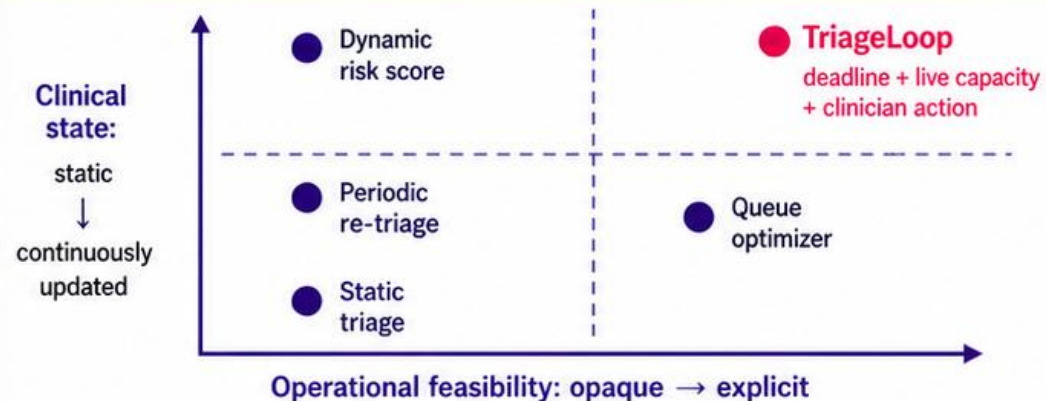
TriageLoop's advantage is the closed loop: eight capabilities connect changing risk to an auditable clinician action.

CAPABILITY COMPARISON | TRIAGELoop CONNECTS ALL 8

8/8

	Static triage	Periodic re-triage	Dynamic risk score	Queue optimizer	Closest published system (2026 India OPD, LLM-based)	TriageLoop
1 Continuous trajectory update	—	✓	✓	—	✓	✓
2 Explicit uncertainty / deferral	—	—	✓	—	—	✓
3 Patient-specific action deadline	—	—	—	—	—	✓
4 Live capacity feasibility	—	—	—	✓	partial	✓
5 No-demotion safety rules	✓	✓	—	—	—	✓
6 Clinician decision receipt	—	—	—	—	—	✓
7 Tamper-evident audit chain	—	—	—	—	—	✓
8 Local outcome feedback	—	—	—	—	—	✓

POSITIONING | CLINICAL DYNAMISM × OPERATIONAL FEASIBILITY



ADVANTAGE COMPOUNDS WITH LOCAL USE



WHY THE CLOSED LOOP IS HARD TO RECREATE



Safety constraints

—
tied to intended use



Beds, staffing and routing

—
configured to the local ED



Override history

—
linked to action receipts



Release decisions

—
linked to measured outcomes

8/8

capabilities in one
accountable loop

Approve the shadow pilot within a \$250k Year-1 cost envelope; production use remains conditional on local safety, nurse workflow and attributable economics.

The proposal funds integration and measurement—not automated patient-care decisions.

APPROVAL REQUIRED

- 1  **Clinical owner**
signs intended use and stop rules
- 2  **De-identified local extract**
supports retrospective replay
- 3  **Integration sandbox**
connects adapter and audit logs
- 4  **Pilot sponsor**
funds the evaluation

MODELLED SINGLE-SITE ECONOMICS



Illustrative scenario—not achieved savings.

PILOT BOUNDARIES

-  Clinicians make every patient-care decision
-  No production use during shadow evaluation
-  Hard-rule outcomes cannot be relaxed
-  Any safety regression triggers stop review

DECISION PACK AT PILOT CLOSE

**CLINICAL SAFETY**

- missed Action Windows
- red-flag sensitivity
- subgroup performance

Owner: Clinical safety

**NURSE WORKFLOW**

- signal-to-review time
- alert burden
- comprehension + overrides

Owner: Triage lead

**ATTRIBUTABLE VALUE**

- boarding minutes
- full operating cost
- payback

Owner: Operations + finance

**TECHNICAL CONTROL**

- data quality
- audit completeness
- failure-recovery tests

Owner: IT + data



Controlled production requires accepted safety, workflow, value and technical-control evidence.

You are not approving a concept. The product is live, tested, and has already survived two rounds of adversarial review — [trialogloop-ai.vercel.app](#).

 **APPROVAL REQUEST: CLINICAL OWNER + LOCAL DATA + SANDBOX ACCESS + PILOT SPONSOR**



Thank you.

triageloop-ai.vercel.app — try the live board.

